Xinyu Shi

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RESEARCH INTERESTS

I am a researcher in **Human-Computer Interaction (HCI)**. I endeavor to answer the question of *how can we exploit the advances in AI automation to augment human creativity and steer creative possibilities.* To achieve this, I design, develop, and evaluate meaningful and useful abstractions, interactions, and tools to enable powerful ways of thinking, visual media authoring, and design intention communication.

EDUCATION

2021 – **Ph.D. in Computer Science**
University of Waterloo, Canada
Advisor: Jian Zhao

2016 – 2020 **B.Eng. in Software Engineering**
Xiamen University, China

PROFESSIONAL EXPERIENCE

Summer 2025 **Adobe Research**, Seattle, WA, USA (Remote)
Research Scientist Intern
Mentors: Rubaiat Habib Kazi, Li-Yi Wei, Cherry Zhao
Proposed a notational animating framework for keyframe authoring.

Summer 2024 **Adobe Research**, San Jose, CA, USA
Research Scientist Intern
Mentors: Shunan Guo, Jane Hoffswell, Gromit Chan, Eunyee Koh
Designed and developed a sketch-based infographics authoring tool.

Summer 2023 **Microsoft Research Asia**, Beijing, China
Research Scientist Intern
Mentor: Yun Wang
Designed and developed a color authoring tool for motion graphics video.

PEER-REVIEWED PUBLICATIONS (FULL PAPERS)

🏆 Best Paper Award; 🎖 Honorable Mention and Other Awards; ∗ Equal Contribution.

Note about venues: In the field of Human-Computer Interaction (HCI), CHI and UIST are considered top tier venues for impactful work, with an annual acceptance rate of 20 - 25%.

[P9] under review — **Xinyu Shi**, Li-Yi Wei, Nanxuan Zhao, Jian Zhao, Rubaiat Habib Kazi. Notational Animating: An Interactive Approach to Creating and Editing Animation Keyframes. *Under Review.*

[P8] under review — **Xinyu Shi**, Chun Ye, Shunan Guo, Jane Hoffswell, Gromit Yeuk-Yin Chan, Eunyee Koh, Jian Zhao. Diagrammatic Sketching: Exploring Infographic Design Alternatives in Parallel. *Under Review.*

[P7] CHI 2025 — **Xinyu Shi**, Yinghou Wang, Ryan Rossi, Jian Zhao. Brickify: Enabling Expressive Design Intent Specification through Direct Manipulation on Design Tokens. *Proc. ACM Human Factors in Computing Systems* (CHI 2025).

[P6] CHI 2024 — **Xinyu Shi**, Yinghou Wang, Yun Wang, Jian Zhao. Piet: Facilitating Color Authoring for Motion Graphics Video. *Proc. ACM Human Factors in Computing Systems* (CHI 2024). 🏆 **Best Paper Award (Top 1%).**

[P5] CHI 2024 — **Xinyu Shi**, Mingyu Liu, Ziqi Zhou, Ali Neshati, Ryan Rossi, Jian Zhao. Exploring Interactive Color Palettes for Abstraction-Driven Exploratory Image Colorization. *Proc. ACM Human Factors in Computing Systems* (CHI 2024).

[P4] CHI 2023 — **Xinyu Shi**, Ziqi Zhou, Julia Zhang, Ali Neshati, Anjul Kumar Tyagi, Ryan Rossi, Shunan Guo, Fan Du, Jian Zhao. De-Stijl: Facilitating Graphics Design with Interactive 2D Color Palette Recommendation. *Proc. ACM Human Factors in Computing Systems* (CHI 2023).

[P3] MM 2022 — Chaofeng Chen*, **Xinyu Shi***, Yipeng Qin, Xiaoming Li, Xiaoguang Han, Tao Yang, Shihui Guo. Real-World Blind Super-Resolution via Feature Matching with Implicit High-Resolution Priors. *Proc. ACM Multimedia* (MM 2022). 🎖 **Oral (Top 5%).**

[P2] Ubicomp 2021 — Qijia Shao, Amy Sniffen, Julien Blanchet, Megan E. Hillis, **Xinyu Shi**, Themistoklis Haris, Jason Liu, Melissa Malzkuhn, Lorna Quandt, James Mahoney, David J.M. Kraemer, Xia Zhou, Devin Balkcom. Teaching American Sign Language in Mixed Reality. *Proc. ACM Interactive, Mobile, Wearable and Ubiquitous Technologies* (IMWUT & Ubicomp 2021).

[P1] ISMAR 2019 — **Xinyu Shi**, Junjun Pan, Zeyong Hu, Juncong Lin, Shihui Guo, Minghong Liao, Ye Pan, Ligang Liu. Accurate and Fast Classification of Foot Gestures for Virtual Locomotion. *Proc. IEEE Intl. Symposium on Mixed and Augmented Reality* (ISMAR 2019).

LIGHTLY PEER-REVIEWED PUBLICATIONS (EXTENDED ABSTRACTS)

[A1] CHI EA 2025 **Xinyu Shi**, Shunan Guo, Jane Hoffswell, Gromit Yeuk-Yin Chan, Victor S. Bursztyn, Jian Zhao, Eunyee Koh. Comprehensive Sketching: Exploring Infographic Design Alternatives in Parallel. *Extended Abstracts of ACM Human Factors in Computing Systems (CHI Late-Breaking Work 2025).*

AWARDS

2024 **Best Paper Award** at ACM CHI 2024 as first author for Piet [P6]

Awarded to the top 1% of submissions made to the CHI conference.

2024 – 2025 **Special Recognitions for Outstanding Reviews** (5 in total)

Recognition to outstanding reviewers provided highly useful peer reviews.

2022 **Oral Recognition** at ACM MM as co-primary author for FeMaSR [P3]

Recognition given to the top 5% of submissions made to the MM conference for oral presentation.

2025 – 2027 **David R. Cheriton Graduate Scholarship**, University of Waterloo

Awarded to top PhD students based on academic research excellence. $20,000 over 2 years.

2022 – 2026 **International Doctoral Student Award**, University of Waterloo

Awarded to top international doctoral students based on academic excellence. $53,000 over 4 years.

2021 **Vector Scholarship in AI**, Vector Institute

Awarded to top incoming graduate students for research potential in AI. $17,500 over 1 year.

2021 **Mathematics International Award of Excellence**, University of Waterloo.

Awarded to incoming international graduate students for research potential. $5,000 over 2 terms.

2016 – 2020 **Award for Excellence in Academic Performance**, Xiamen University

Awarded to top undergraduate students based on academic excellence. 5,000 CNY each time.

PAST RESEARCH POSITIONS

2021/09 – Present **University of Waterloo**, Waterloo, ON, Canada
Graduate Student Researcher
Advisor: Jian Zhao
Building interactive systems to augment human creativity [P4, P5, P7].

2020/12 – 2021/08 **The Chinese University of Hong Kong, Shenzhen**, Shenzhen, China
Research Assistant
Advisor: Xiaoguang Han
Proposed an approach based on VQ-GAN for Image Super-Resolution [P3].

2019/09 – 2020/03 **Dartmouth College**, Hanover, NH, USA
Research Assistant
Advisors: Xia Zhou, Xing-Dong Yang
Leveraged wearable devices & Mixed Reality for Sign-Language teaching [P2].

2017/10 – 2019/09 **Xiamen University**, Xiamen, China
Undergraduate Research Assistant
Advisor: Shihui Guo
Exploited wearable devices for VR locomotion with gesture recognition [P1].

ACADEMIC SERVICES

Organizing Committee

2025 – 2026 Paper Chair Assistant, CHI 2026

Program Committee

2025 International Program Committee, GI 2025

2025 Program Committee, Associate Chair, C&C 2025

Conference Reviewer

2024 – 2025 SIGCHI Conference on Human Factors in Computing Systems (CHI)

2024 – 2025 ACM User Interface Software and Technology Symposium (UIST)

2024 – 2025 IEEE Visualization and Visual Analytics Conference (VIS)

2024 – 2025 SIGCHI Late Breaking Work (CHI LBW)

2025 Pacific Conference on Computer Graphics and Applications (Pacific Graphics)

2024 ACM Conference on Designing Interactive Systems (DIS)

2024 Creativity & Cognition (C&C)

Special Recognitions for Outstanding Reviews ($\times 5$):
CHI'24, CHI'25, UIST'25, CHI LBW'25, C&C'25

Conference Session Chair

2025 Paper Session "*Malleable and Adaptive Interface*", CHI 2025

INVITED TALKS

Interactive Visual Abstractions for Steerable Creativity

2025/11 University of Toronto iSchool Seminar, hosted by Prof. Ebrahim Bagheri.

2025/11 Guest Lecture for CS889 in University of Waterloo, hosted by Prof. Jian Zhao.

2025/11 UCSD Design Lab Seminar

MENTORING

2025 – **Connie Cheng**, Master student at Hard University & MIT.
Project: Specifying temporal change for animated brush.

2025 – **Cara Li**, Master student at University of Waterloo.
Project: Fluid in-situ graphics creation through reified sketched annotations.

2024 – **Yue Theresa Lyu**, Ph.D. student at University of Waterloo.
Project: video creation and authoring tool.

2024 – 2025 **Chun Daisy Ye**, Undergraduate student at University of Waterloo.
Now: Software Engineer at Salesforce.

2023 – 2024 **Yinghou Wang**, Master student at Harvard University & MIT, published [P6, P7].
Now: Creative Technologist at Motorola Solutions.

2023 **Mingyu Liu**, Undergraduate student at University of Waterloo, published [P5].
Now: Software Engineer at Bloomberg.

2022 **Ziqi Zhou**, Master student at University of Waterloo, published [P4].
Now: Partner at Puzzle Ventures.

2022 **Julia Zhang**, Undergraduate student at University of Waterloo, published [P4].
Now: Software Engineer at Whatnot.

TEACHING EXPERIENCE

Teaching Assistant in University of Waterloo

CS246 Object-Oriented Software Development. (Winter 2022, Fall 2023, Winter 2025)

CS251 Computer Organization and Design. (Spring 2022, Winter 2023)

CS234 Data Types and Structures. (Fall 2022)

CS115 Introduction to Computer Science. (Fall 2021)

PRESS COVERAGE

2024 *Waterloo News*: Scheming colours faster and easier. [Link]

2023 *GraphicSpeak*: University of Waterloo's Computer Scientists want Graphic Designers to use better Colors. [Link]

2023 *TechTimes*: Computer Scientists Develop "De-Stijl" Tool With Adobe to Help People Use Color Better in Graphic Design. [Link]

2023 *Waterloo News*: Dreaming in technicolour: Waterloo computer scientists develop a tool to help people use colour better in graphic design. [Link]